11 Chemistry — Unit 2 Revision Worksheet: worked solutions

WORKED SOLUTIONS — attempt every question before you check

87 marks across 30 questions. Each **1 mark** shows one step or idea that would earn a mark in an examination. *Consequential:* means you can still earn the mark if you used your own (wrong) earlier answer correctly. *Reject:* lists common answers that would not earn the mark. Data: V_m at SLC = 24.8 L mol^{-1}; R = 8.31 J mol^{-1} K^{-1}; c(water) = 4.18 J g^{-1} $°C^{-1}$; 0 °C = 273 K; 1 atm = 101.3 kPa.

Topic 1 – Solution concentration and unit conversions

Question 1 — Easy (2 marks)

a) (1 mark)

1 mark 30.0 g ÷ 0.500 L = 60.0 g L^{-1}.

Reject: 15.0 g L^{-1} (multiplied by 0.500).

b) (1 mark)

1 mark % (m/v) = grams per 100 mL: 30.0 g per 500 mL = 6.00 g per 100 mL = 6.00% (m/v).

Reject: 60.0% (used g per L).

Question 2 — Medium (2 marks)

Answer (2 marks)

1 mark n = 0.0400 × 0.250 = 0.0100 mol. **1 mark** M($MgSO_4$) = 24.3 + 32.1 + 4 × 16.0 = 120.4 g mol^{-1}; m = 0.0100 × 120.4 = 1.20 g.

Reject: Using 250 instead of 0.250 L (1204 g).

Question 3 — Hard (4 marks)

a) (2 marks)

1 mark 0.0250 g per 100 mL = 0.250 g L^{-1}. **1 mark** 0.250 ÷ 159.6 = 1.57×10^{-3} M.

Reject: Using 0.0250 g L^{-1} (forgot to multiply by 10).

b) (2 marks)

1 mark c(Cu^{2+}) = c($CuSO_4$) = 1.566×10^{-3} mol L^{-1}; mass = $1.566 \times 10^{-3} \times 63.5 = 0.0995$ g L^{-1}. **1 mark** = 99.5 mg L^{-1}.

Consequential: Consequential on part a).

Reject: 250 mg L^{-1} (the mass of $CuSO_4$, not Cu^{2+}).

Topic 2 – Standard solutions and dilution

Question 4 — Easy (1 mark)

Answer (1 mark)

1 mark $V_1 = c_2V_2 \div c_1 = 0.0640 \times 250.0 \div 0.800 = 20.0$ mL.

Reject: 3125 mL (concentrations inverted).

Question 5 — Medium (3 marks)

a) (2 marks)

1 mark n = 0.0750 × 0.1000 = 7.50×10^{-3} mol. **1 mark** m = $7.50 \times 10^{-3} \times 204.2 = 1.53$ g.

Reject: Using 100.0 instead of 0.1000 L (1530 g).

b) (1 mark)

1 mark Any one: available in very high purity; stable (does not react with air or absorb water from the air); high molar mass (so weighing errors are small); soluble in water.

Reject: "It is an acid." "It is cheap."

Question 6 — Hard (3 marks)

a) (2 marks)

1 mark after Step 1: 0.500 × 20.00 ÷ 500.0 = 0.0200 M. **1 mark** after Step 2: 0.0200 × 5.00 ÷ 100.0 = 1.00×10^{-3} M (overall dilution factor 25 × 20 = 500).

Reject: Adding the dilution factors (25 + 20) instead of multiplying them.

b) (1 mark)

1 mark $1.00 \times 10^{-3} \times 158.2 = 0.158$ g L^{-1} = 158 mg L^{-1}.

Consequential: Consequential on part a).

Reject: Leaving the answer in g L^{-1}.

Topic 3 – Volumetric analysis

Question 7 — Easy (2 marks)

a) (1 mark)

1 mark 25.20 − 1.35 = 23.85 mL.

Reject: 26.55 mL (added the readings).

b) (1 mark)

1 mark an aliquot.

Reject: Titre (that is the volume delivered from the burette).

Question 8 — Medium (3 marks)

Answer (3 marks)

1 mark n(HCl) = 0.0800 × 0.01460 = 1.168×10^{-3} mol; $n(Ca(OH)_2) = ½ \times 1.168 \times 10^{-3} = 5.84 \times 10^{-4}$ mol. **1 mark** c = $5.84 \times 10^{-4} \div 0.02500 = 0.0234$ M. **1 mark** $M(Ca(OH)_2)$ = 74.1 g mol^{-1}; 0.02336 × 74.1 = 1.73 g L^{-1}.

Reject: 1:1 ratio (3.46 g L^{-1}). Doubling instead of halving (6.92 g L^{-1}).

Question 9 — Hard (6 marks)

a) (2 marks)

1 mark n(HCl) in a 20 mL titre = 0.100 × 0.0200 = 2.00×10^{-3} mol = n(KOH) in the aliquot. **1 mark** c(KOH) = $2.00 \times 10^{-3} \div 0.02500 = 0.0800$ M.

Reject: Answers that use 2.5 M directly.

b) (2 marks)

1 mark dilution factor = 2.5 ÷ 0.0800 ≈ 31. **1 mark** any workable plan with volumetric glassware giving a titre within the 50.00 mL burette, e.g. pipette 10.00 mL of the KOH solution into a 250.0 mL volumetric flask and make it up to the mark (1 in 25, titre ≈ 25 mL), or 5.00 mL made up to 250.0 mL (1 in 50, titre ≈ 12.5 mL).

Consequential: Consequential on part a).

c) (2 marks)

1 mark each 25.00 mL aliquot of 2.5 M KOH contains 0.0625 mol of KOH, which would need 0.0625 ÷ 0.100 = 0.625 L (625 mL) of HCl. **1 mark** this is far more than the 50.00 mL burette holds, so the end point could not be reached in one filling.

Reject: “It is too concentrated” with no link to the titre volume (max 1).

Topic 4 – Greenhouse gases

Question 10 — Easy (2 marks)

Answer (2 marks)

1 mark CH_4: e.g. livestock farming (cattle, sheep), rice paddies, landfill, leaks from gas wells or pipelines. **1 mark** CO_2: e.g. burning coal, oil, petrol or natural gas; clearing and burning forests; cement making.

Reject: The same activity for both gases unless it genuinely adds both and is explained.

Question 11 — Medium (2 marks)

Answer (2 marks)

1 mark the warm surface emits infrared radiation; CO_2 molecules absorb some of this infrared radiation (they let most visible light through). **1 mark** the molecules then re-emit the energy in all directions, including back towards the surface, so less energy escapes to space and the lower atmosphere warms.

Reject: “CO_2 reflects sunlight.” “CO_2 absorbs ultraviolet light.”

Question 12 — Hard (4 marks)

a) (2 marks)

1 mark CO_2: (410 − 337) ÷ 337 × 100 = 21.7%. **1 mark** CH_4: (1.83 − 1.55) ÷ 1.55 × 100 = 18.1%.

Reject: Dividing by the 2020 value.

b) (2 marks)

1 mark molecule for molecule, methane has a much stronger warming effect than CO_2 (roughly 25–30 times over a century). **1 mark** so even a small rise in methane concentration adds noticeably to the enhanced greenhouse effect.

Reject: “Methane is more concentrated” (it is not). Answers about methane being flammable.

Topic 5 – Gas pressure, SLC and pV = nRT

Question 13 — Easy (2 marks)

a) (1 mark)

1 mark 1.50 × 101.3 = 152 kPa.

Reject: 1.48 kPa (divided).

b) (1 mark)

1 mark 45.0 + 273 = 318 K.

Reject: 228 K (subtracted).

Question 14 — Medium (3 marks)

a) (1 mark)

1 mark Total volume = 12.00 L; p_2 = 100 × 3.00 ÷ 12.00 = 25.0 kPa (the amount and temperature are unchanged, so pV is constant).

Reject: 33.3 kPa (used 9.00 L as the final volume).

b) (2 marks)

1 mark the same number of particles, moving at the same average speed, now spread through a larger volume. **1 mark** so fewer particles strike each part of the walls per second, and the pressure is lower.

Reject: “The particles slow down” (the temperature is constant). “Some gas is lost.”

Question 15 — Hard (4 marks)

a) (2 marks)

1 mark T = −40.0 + 273 = 233 K. **1 mark** V = nRT ÷ p = 30.0 × 8.31 × 233 ÷ 30.0 = 1.94×10^3 L (accept 1935–1940 L).

Reject: Using −40.0 for T (negative volume). Using 101 kPa.

b) (2 marks)

1 mark the pressure falls from 101 kPa to 30.0 kPa (to about 0.30 of its value), while the kelvin temperature falls only from 290 K to 233 K (to about 0.80 of its value). **1 mark** volume is proportional to T ÷ p, so the drop in pressure has the larger effect and the volume increases (by about 2.7 times).

Reject: Answers that compare the temperatures in °C. “Helium is lighter than air” (explains rising, not expanding).

Topic 6 – Molar volume and molar mass of gases

Question 16 — Easy (2 marks)

a) (1 mark)

1 mark V_m = V ÷ n = 0.251 ÷ 0.0100 = 25.1 L mol^{-1}.

Reject: 251 ÷ 0.0100 = 25 100 (mL not converted to L).

b) (1 mark)

1 mark (25.1 − 24.8) ÷ 24.8 × 100 = 1.2% higher.

Reject: Answers with no reference value stated.

Question 17 — Medium (3 marks)

a) (2 marks)

1 mark n = pV ÷ RT = 105 × 0.401 ÷ (8.31 × 313) = 0.01619 mol. **1 mark** M = 0.712 ÷ 0.01619 = 44.0 g mol^{-1}.

Reject: V = 401 (mL not converted). Using 24.8 L mol^{-1} (the gas is not at SLC).

b) (1 mark)

1 mark carbon dioxide, CO_2 (M = 44.0 g mol^{-1}).

Consequential: Consequential on part a).

Question 18 — Hard (3 marks)

Answer (3 marks)

1 mark n(CO_2) = 5.00 ÷ 24.8 = 0.2016 mol. **1 mark** n($C_6H_{12}O_6$) = ½ × 0.2016 = 0.1008 mol. **1 mark** m = 0.1008 × 180.0 = 18.1 g.

Reject: Multiplying by 2 instead of halving (72.6 g). 1:1 ratio (36.3 g).

Topic 7 – Limiting and excess reactants

Question 19 — Easy (2 marks)

Answer (2 marks)

1 mark 2.0 mol N_2 needs 6.0 mol H_2; 9.0 mol is present, so N_2 is limiting. **1 mark** $n(NH_3) = 2 \times 2.0 = 4.0$ mol.

Reject: 6.0 mol NH_3 (calculated from H_2, the excess reactant).

Question 20 — Medium (3 marks)

Answer (3 marks)

1 mark n(Cu) = 6.35 ÷ 63.5 = 0.100 mol; n(S) = 1.28 ÷ 32.1 = 0.0399 mol. **1 mark** 0.100 mol Cu needs 0.0500 mol S, but only 0.0399 mol is present, so sulfur is limiting. **1 mark** $n(Cu_2S) = 0.0399$ mol; m = 0.0399 × 159.1 = 6.34 g.

Reject: Treating the ratio as 1:1 (copper limiting). Adding the masses (7.63 g).

Question 21 — Hard (4 marks)

a) (3 marks)

1 mark n(Zn) = 6.54 ÷ 65.4 = 0.100 mol; $n(I_2) = 20.0 \div 253.8 = 0.0788$ mol. **1 mark** I_2 is limiting (1:1 ratio). **1 mark** $m(ZnI_2) = 0.0788 \times 319.2 = 25.2$ g.

Reject: Using M(I) = 126.9 for iodine (0.158 mol, wrongly making Zn limiting).

b) (1 mark)

1 mark n(Zn) left = 0.100 − 0.0788 = 0.0212 mol; m = 0.0212 × 65.4 = 1.39 g.

Consequential: Consequential on part a).

Topic 8 – Specific heat capacity and calorimetry

Question 22 — Easy (2 marks)

a) (1 mark)

1 mark q = 1200 × 4.18 × 50.0 = 250 800 J = 251 kJ.

Reject: Using 1.20 (kg not converted to g): 0.251 kJ. Using 65.0 °C instead of ΔT.

b) (1 mark)

1 mark Greater, by a factor of 4.18 ÷ 2.0 ≈ 2.1 (ΔT ≈ 105 °C), because the oil needs less energy per gram per °C.

Reject: “Smaller, because oil is thicker.”

Question 23 — Medium (3 marks)

a) (1 mark)

1 mark q = 100.0 × 4.18 × (22.0 − 16.1) = 2.47×10^3 J (2466 J). Using the solution mass, 108.0 g, is also accepted (2.66×10^3 J).

Reject: A negative value for the energy absorbed.

b) (2 marks)

1 mark n = 8.00 ÷ 80.0 = 0.100 mol. **1 mark** 2.466 kJ ÷ 0.100 = 24.7 kJ mol^{-1}.

Consequential: Consequential on part a).

Question 24 — Hard (5 marks)

a) (3 marks)

1 mark energy lost by the hot water = energy gained by the cold water. **1 mark** $200 \times 4.18 \times (80.0 - T) = 300 \times 4.18 \times (T - 20.0)$, so $16\,000 - 200T = 300T - 6000$. **1 mark** $500T = 22\,000$, $T = 44.0$ °C.

Reject: Averaging the temperatures (50.0 °C, ignores the different masses).

b) (2 marks)

1 mark energy lost by the hot water = $200 \times 4.18 \times (80.0 - 41.0) = 32\,604$ J; energy gained by the cold water = $300 \times 4.18 \times (41.0 - 20.0) = 26\,334$ J. **1 mark** difference = 6270 J = 6.27 kJ (transferred to the cup and the surroundings).

Reject: Using 44.0 °C in either term.

Topic 9 – Redox: oxidising agents, reducing agents and equations

Question 25 — Easy (2 marks)

a) (1 mark)

1 mark sodium, Na ($0 \rightarrow +1$, loses electrons).

Reject: Cl_2. NaCl.

b) (1 mark)

1 mark chlorine, Cl_2 (it gains electrons and is reduced).

Reject: Na.

Question 26 — Medium (2 marks)

Answer (2 marks)

1 mark Cr and O balanced: $Cr_2O_7^{2-} + 14H^+ \rightarrow 2Cr^{3+} + 7H_2O$. **1 mark** $6e^-$ on the left, with states: $Cr_2O_7^{2-}(aq) + 14H^+(aq) + 6e^- \rightarrow 2Cr^{3+}(aq) + 7H_2O(l)$.

Reject: $3e^-$ (forgot there are two Cr atoms). Balancing with OH^-.

Question 27 — Hard (3 marks)

a) (1 mark)

1 mark $2I^-(aq) \rightarrow I_2(aq) + 2e^-$.

Reject: $I^- \rightarrow I + e^-$ (iodine is diatomic).

b) (1 mark)

1 mark $H_2O_2(aq) + 2H^+(aq) + 2e^- \rightarrow 2H_2O(l)$.

Reject: $H_2O_2 \rightarrow H_2O + O$ (not balanced with H^+ and e^-).

c) (1 mark)

1 mark $H_2O_2(aq) + 2H^+(aq) + 2I^-(aq) \rightarrow I_2(aq) + 2H_2O(l)$.

Consequential: Consequential on parts a) and b).

Reject: Electrons shown in the overall equation.

Topic 10 – Reactivity series, corrosion and primary cells

Question 28 — Easy (3 marks)

a) (1 mark)

1 mark Any one: grey/silver crystals form on the wire; the solution turns blue.

Reject: “Bubbles of gas.” “The copper glows.”

b) (2 marks)

1 mark $Cu + 2Ag^+ \rightarrow Cu^{2+} + 2Ag$ (balanced for charge). **1 mark** states: $Cu(s) + 2Ag^+(aq) \rightarrow Cu^{2+}(aq) + 2Ag(s)$.

Reject: $Cu + Ag^+ \rightarrow Cu^{2+} + Ag$ (charge not balanced).

Question 29 — Medium (3 marks)

a) (1 mark)

1 mark silver oxide, Ag_2O.

Reject: Zn. Ag.

b) (1 mark)

1 mark $Zn(s) + Ag_2O(s) \rightarrow ZnO(s) + 2Ag(s)$ (H_2O and OH^- cancel).

Reject: Equations that still show electrons or OH^- on both sides.

c) (1 mark)

1 mark $n(Zn) = ½ \times 0.0100 = 5.00 \times 10^{-3}$ mol; $m = 5.00 \times 10^{-3} \times 65.4 = 0.327$ g.

Reject: 0.654 g (forgot the 2 electrons per Zn).

Question 30 — Hard (4 marks)

a) (1 mark)

1 mark oxygen, O_2.

Reject: Fe. Water.

b) (3 marks)

1 mark $V(O_2)$ used = 100.0 − 79.5 = 20.5 mL; $n(O_2) = 0.0205 \div 24.8 = 8.27 \times 10^{-4}$ mol. **1 mark** $n(Fe) = 4/3 \times 8.27 \times 10^{-4} = 1.10 \times 10^{-3}$ mol. **1 mark** $m = 1.10 \times 10^{-3} \times 55.8 = 0.0615$ g (accept 0.0614–0.0616 g).

Reject: Using 79.5 mL as the volume of oxygen. Ratio 3/4 instead of 4/3 (0.0346 g).

Mark summary

Topic	Easy	Medium	Hard
Solution concentration and unit conversions	2	2	4
Standard solutions and dilution	1	3	3
Volumetric analysis	2	3	6
Greenhouse gases	2	2	4
Gas pressure, SLC and pV = nRT	2	3	4
Molar volume and molar mass of gases	2	3	3
Limiting and excess reactants	2	3	4
Specific heat capacity and calorimetry	2	3	5
Redox: oxidising agents, reducing agents and equations	2	2	3
Reactivity series, corrosion and primary cells	3	3	4
Total	20	27	40